Sec: SR.IIT _*(MODEL-A) **Date: 06-01-26**
Time: 3 Hrs **SPL GTM TEST** **Max. Marks: 300**

KEY SHEET
PHYSICS

1	**D**	2	**A**	3	**C**	4	**D**	5	**B**
6	**A**	7	**A**	8	**A**	9	**B**	10	**C**
11	**A**	12	**A**	13	**B**	14	**D**	15	**D**
16	**D**	17	**B**	18	**B**	19	**D**	20	**A**
21	**7**	22	**2**	23	**4**	24	**2**	25	**4**

CHEMISTRY

26	**D**	27	**D**	28	**A**	29	**B**	30	**C**
31	**B**	32	**B**	33	**B**	34	**B**	35	**A**
36	**D**	37	**C**	38	**D**	39	**B**	40	**D**
41	**C**	42	**D**	43	**C**	44	**A**	45	**B**
46	**25**	47	**6**	48	**2**	49	**5**	50	**333**

MATHEMATICS

51	**D**	52	**B**	53	**A**	54	**B**	55	**C**
56	**C**	57	**C**	58	**A**	59	**A**	60	**B**
61	**D**	62	**C**	63	**B**	64	**B**	65	**A**
66	**C**	67	**A**	68	**B**	69	**C**	70	**A**
71	**4**	72	**499**	73	**2**	74	**7**	75	**1**

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PHYSICS

1.Ans (D)

Since external force acting on the system in horizontal direction is zero & $V_{com} = 0$ so

2. Ans (A)

Closer the readings, greater the precision.

3. Ans (C)

Conceptual

4. Ans (D)

$A_1 Y_1 \alpha_1 = A_2 Y_2 \alpha_2$

5. Ans (B)

I α width

So $I_\alpha = I_1$

$I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 = 9I_1$

$I_{min} = \left(\sqrt{I_2} - I_1\right)^2 = I_1$

Fringe visibility $= \dfrac{I_{max} - I_{min}}{I_{max} + I_{min}}$

$= \dfrac{SI_1 - I_1}{SI_1 + I_1} \times 100$

$= \dfrac{8}{10} \times 100 = 80\%$

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6. Ans (A)

Get μ, λ and τ be units mass length and time in the new system. Now energy is $\frac{1}{2}mv^2$ i.e. it units will be $\mu\lambda^2/\tau^2$. Thus our unit energy becomes

$5\ kg \times (20\ m)^2 = 20k_5\ m^2/sec^2 = 20$ Joule.

Thus connect option is A .

7. Ans (A)

I_{rmo} for riding part

$= \dfrac{10}{\sqrt{3}}$ A

For flat part $= 10$ A

$$I_{rmo} = \left(\frac{\left(\frac{10}{\sqrt{3}}\right)^2 \times 2 \times (10)^2 \times 2}{4} \right)^{1/2}$$

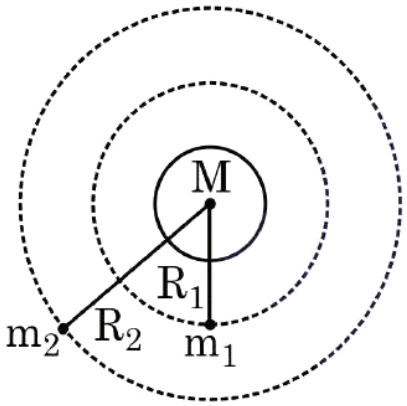
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$$= \left(\frac{800}{3\times 4}\right)^{1/2} = \left(\frac{200}{3}\right)^{1/2} = 8.16 \text{ A}$$

8. Ans (A)

Conceptual

9. Ans (B)



$$\left.\begin{aligned}\frac{m_1}{m_2} &= 2\\ \frac{R_1}{R_2} &= \frac{1}{4}\end{aligned}\right\} \text{ given}$$

$$\frac{GMm_1}{R_1^2} = \frac{m_1 v_1^2}{R_1}$$

$$v_1^2 = \frac{GM}{R_1}, v_2^2 = \frac{GM}{R_2}$$

$$\frac{v_1^2}{v_2^2} = \frac{R_2}{R_1} = 4$$

(P) $\frac{V_1}{V_2} = 2$

(Q) $L = mvR$

$$\frac{L_1}{L_2} = \frac{m_1 v_1 R_1}{m_2 v_2 R_2} = 2\times 2\times \frac{1}{4} = 1$$

(R) $K = \frac{1}{2}mv^2$

$$\frac{K_1}{K_2} = \frac{m_1 v_1^2}{m_2 v_2^2} = 2\times (2)^2 = 8$$

(S) $T = 2\pi R / V$

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$$\frac{T_1}{T_2}=\frac{R_1}{v_1}\times\frac{v_2}{R_2}=\frac{R_1}{R_2}\times\frac{v_2}{v_1}=\frac{1}{4}\times\frac{1}{2}=\frac{1}{8}$$

10. Ans (C)

$$\vec{E}=-\frac{dV}{dx}\hat{i}$$

$$\vec{E}=-\frac{d}{dx}\left[2x^2-4x\right]\hat{i}$$

11. Ans (A)

Correct answer is (A)

12. Ans (A)

$$R_{eq}=\frac{600}{3}+200=400\Omega$$

$$i=\frac{\varepsilon}{R_{eq}}=\frac{200\text{ V}}{400\Omega}=0.5\text{ A}\ \text{(Reading of (A))}$$

Reading of V = TPD of cell

$$=200-(0.5)(200)$$

$$=100\text{ V}$$

13. Ans (B)

Correct answer is (B)

14. Ans (D)

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Correct answer is (D)

15. Ans (D)

Theoretical

16. Ans (D)

$$A=A_0e^{-bt/2m}$$

$$\Rightarrow\frac{A_0}{3}=A_0e^{\frac{-b}{2m}(100\text{ T})}\quad T:\text{Time period}$$

$$A'=A_0e^{\frac{-b}{2m}(200\text{ T})}$$

$$\Rightarrow A'=A_0\left(\frac{1}{3}\right)^2$$

17. Ans (B)

$$I=\frac{1}{2}\varepsilon_0E_0^2c$$

$$I=\frac{1}{2}\times\left(8.85\times10^{-12}\right)(56.5)^2\times\left(3\times10^8\right)$$

$$=4.24\text{Wm}^{-2}.$$

18. Ans (B)

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$$\frac{\omega_A}{f_A}+\frac{\omega_B}{f_B}=0$$

$$\Rightarrow \frac{f_A}{f_B}=-\frac{\omega_A}{\omega_B}=-\frac{2}{3}$$

$$\Rightarrow f_A=-\frac{2}{3}f_B$$

$$\Rightarrow \frac{1}{f_A}+\frac{1}{f_B}=P_{eq}=-0.8D-\frac{3}{2f_B}+\frac{1}{f_B}=-\frac{0.8}{100}$$

$$-\frac{1}{2f_B}=-\frac{0.8}{100}$$

$$f_B=\frac{100}{1.6}\text{ cm}=62.5\text{ cm}$$

19. Ans (D)

Correct answer is (D)

20. Ans (A)

The magnetic field at point P due to wire of length '13a' and '5a' is zero. Because ' P ' lies on the wire.

Magnetic field at point P due to wire of length '12a'

$$B=\frac{\mu_0 i}{4\pi 5a}\left[\sin\theta_1+\sin 0^\circ\right],\otimes\#(\otimes)$$

$$B=\frac{\mu_0 i}{4\pi 5a}\left[\frac{12a}{13a}\right],$$

$$B=\frac{3\mu_0 i}{65\pi a}\otimes$$

21. Ans (7)

$$\frac{1}{2}mv^2=\frac{\pi R^2}{2}+\frac{\pi}{2}\left(\frac{R}{2}\right)^2+\pi\left(\frac{R}{4}\right)^2+\ldots.$$

$$=\frac{\pi R^2}{2}\left[1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\ldots\ldots\right]$$

$$=\frac{\pi R^2}{2}\times\frac{1}{1-\frac{1}{4}}$$

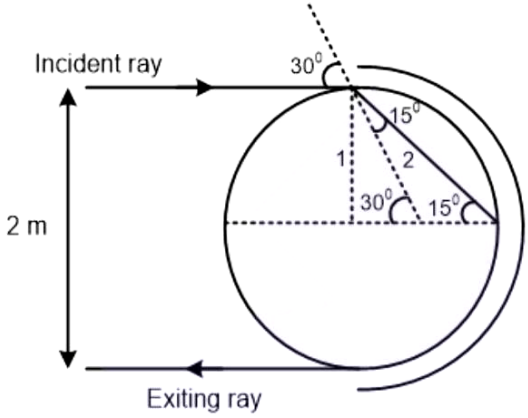
$$\frac{1}{2}\pi v^2=\frac{4\pi R^2}{6}$$

$$v=\sqrt{\frac{4\pi F_0^2}{3m}}$$

22. Ans (2)

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Applying Snell's law at the spherical refracting surface, we have

$\sin 30^0 = \mu \sin 15^0$

$2\sin 15^0 \cdot \cos 15^0 = \mu \sin 15^0$

$\mu = 2\cos 15^0$

23. Ans (4)

Correct answer is (4)

24. Ans (2)

From graph young's Modulus = slope

wire-1, $\gamma_1 = \frac{20}{5} = 4\text{kN/m}^2$

wire-2, $\gamma_2 = \frac{20}{20} = 1\text{kN/m}^2$

Now, strain $= \frac{\text{stress}}{\gamma} \Rightarrow \frac{\Delta L_1}{L_1} = \frac{40}{4} = 10$

$\frac{\Delta L_2}{L_2} = \frac{10}{1} = 10$

$\frac{\Delta L_1}{\Delta L_2} = \frac{10\, L_1}{10\, L_2} = \frac{L_1}{L_2} = \frac{2}{1}$

25. Ans (4)

$\because 2x^2 - 8x + 3y = 0, \therefore y = \frac{8}{3}x - \frac{2x^2}{3}$

$\therefore y = \frac{8}{3}x\left(1 - \frac{x}{4}\right)$

by comparing it with $y = x\tan\theta\left(1 - \frac{x}{R}\right)$

$\therefore R = 4$ m

CHEMISTRY

26. Ans (D)

Cation (Na^+) is formed by loss of 1 electron. More effective nuclear charge, more attraction Radius decreases

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27. Ans (D)

For 2s orbital

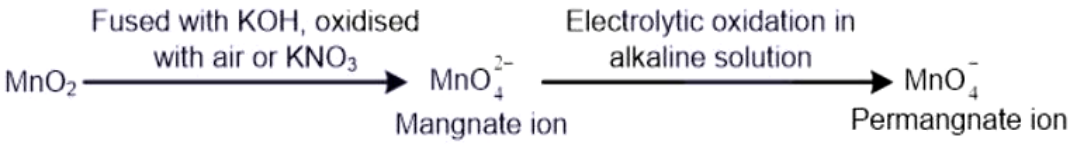
$$\text{radial node} = n - \ell - 1$$
$$= 2 - 0 - 1$$
$$= 1$$

That means curve starts at a positive value near the nucleus, rises to a peak drops to zero at a radial node, then rises to a second peak and finally decreases towards zero.

28. Ans (A)

Stability of +5 O.S. state decreases and that of +3 oxidation state increase down the group in group 15. Inert pair effect is more prone in case of heavier members of p-block, due to increase in size as we move down the group.

29. Ans (B)



30. Ans (C)

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$$\Lambda_m^c = \frac{K \times 1000}{\text{Molarity}}$$
$$= \frac{10^{-5} \times 1000}{0.01} = 1 \text{ S cm}^2 \text{ mol}^{-1}$$

Degree of dissociation (α)

$$= \frac{\Lambda_m^c}{\Lambda_m^o} = \frac{1}{100}$$
$$\alpha = 0.01$$

Percentage dissociation $= 1\%$

31. Ans (B)

(A) $\left[H^+\right] = 1 \times 10^{-7} = c\alpha$

$$\alpha = \frac{10^{-7}}{55.5} \left[\text{Molarity}(c) = 55.5M \right]$$
$$1.8 \times 10^{-9}$$

(B) $Pk_a + Pk_b = 14$

(C) $CH_4 < NH_3 < H_2O < HF$ (Acidic)

with increase in Electronegativity Acidic strengh increases

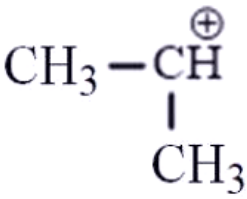
(D) Due to common ion effect, solubility decreases.

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32. Ans (B)

(i) Sodium reduces $H^{\oplus}$ to H_2(g)

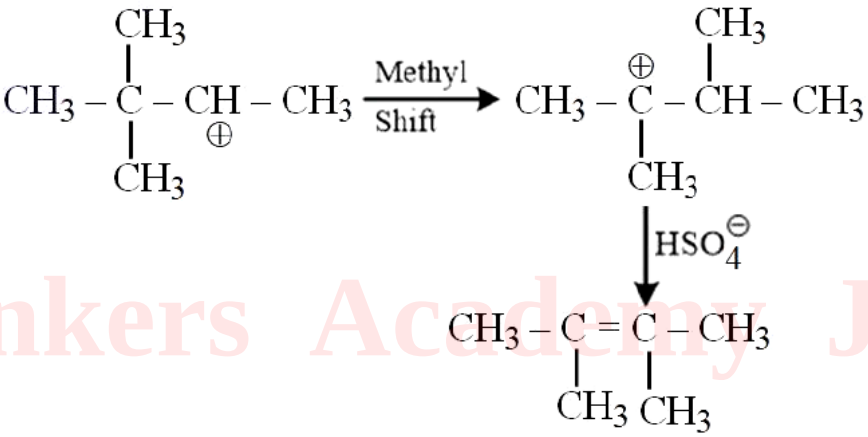
(ii) Due to formation of stable



Carbocation.

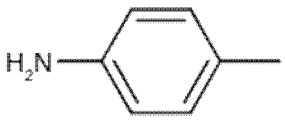
(iii) Antimarkovnikov product obtain.

(iv)



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33. Ans (B)



34. Ans (B)

Sucrose → Cane sugar, Lactose → Milk sugar, Glycogen → Animal starch, Starch → Cereals, roots, tubers

35. Ans (A)

Nitriles are reduced to corresponding imine with stannous chloride in the presence of hydrochloric acid, which on hydrolysis give corresponding aldehyde.

$RCN + SnCl_2 + HCl$

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$$RCH=NH \xrightarrow{H_3O^+} RCHO$$

36. Ans (D)

(A) IE_1 of Be is more than Boron as an S electron is attracted to the nucleus is more than p-electron in same principle quantum no.

(C) $Be(OH)_2$ is Amphoteric so, soluble in aquous NaoH.

37. Ans (C)

It is important to note that double bond in C_2 consists of both pi bonds because of the presence of four electrons in two pi molecular orbitals.

$\sigma 1s^2\sigma^* 1s^2\sigma 2s^2\sigma^* 2s^2\pi 2p_x^2 = \pi 2p_y^2$

38. Ans (D)

$2NaI + 2H_2SO_4 \rightarrow Na_2SO_4 + SO_2 + 2H_2O + I_2$

I_2 + Starch solution → Blue colour

$NaI + H_2SO_4 \rightarrow NaHSO_4 + HI$

$2HI + H_2SO_4 \rightarrow 2H_2O + I_2 + SO_2$

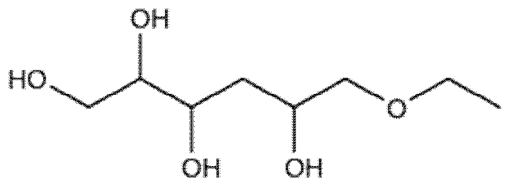
$6NaI + 4H_2SO_4 \rightarrow 3I_2 + 4H_2O + S + 3Na_2SO_4$

$8NaI + 5H_2SO_4 \rightarrow 4I_2 + H_2S + 4Na_2SO_4 + 4H_2O$

39. Ans (B)

(II) Order of boiling point

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(III) Methanol is more poisonous than ethanol.

(IV) As molar mass of 1° alcohol increases its solubility decreases.

40. Ans (D)

- One should read lower meniscus in case of colourless and transparent solution and upper meniscus for solutions of dark colors.
- A secondary standard can not be used for preparing standard solution by direct weighing
  (NCERT Lab Manual)

41. Ans (C)

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In partition chromatography, stationary phase is thin film of liquid adsorbed on an essentially inert support. Mobile phase may be a liquid or a gas. Paper chromatography is an example of partition chromatography in which liquid present in the pores of paper is stationary phase and some other liquid is phases and the adsorption effects of inert support on compounds undergoing chromatographic separation.

42. Ans (D)

Cationic part of solid XeF_6 is XeF_5^+ having sp^3d^2 hybridisation.

C_2, Li_2, O_2^{2+} have no unpaired electron so, these are diamagnetic in nature.

$MgO < Al_2O_3 < SiO_2$ (acidic nature) on moving from left to right non metallic character increases so acidic strength increases. Shape of IF_7 is pentagonal bipyrimidal (having steric number 7)

43. Ans (C)

The M-C π bond is formed by the donation of a pair of electrons from a filled d orbital of metal into the vacant antibonding π^* orbital of carbon monoxide.

44. Ans (A)

$W = -30 = -nR\Delta T$

$\Delta U = nC_V\Delta T$

$= n\left(\frac{6}{2}\times R\right).\Delta T = 90\ J$

$Q = \Delta U - W$

$= 90 - (-30)$

$= 120\ J$

45. . Ans (B)

On increasing the temperature K_{eq} increases, hence the reaction is Endothermic.

46. Ans (25)

Catalyst does not alter enthalpy of reaction (ΔH).

47. Ans (6)

a = Total no. of compounds show positive tollens test:- (I), (IV), (VII)

both alphatic & aromatic aldehyde show positive tollens test

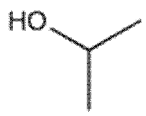
b = Total no. of compounds show positive

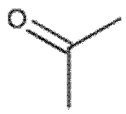
haloform test :- (II), (III),(VI)

compound having

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groups show positive haloform test

$a = 3$

$b = 3$

48. Ans (2)

$$R-\overset{O}{C}-NH_2 + Br_2 + 4NaOH \rightarrow$$

$$RNH_2 + Na_2CO_3 + 2NaBr + 2H_2O$$

49. Ans (5)

$$\ln\frac{K_2}{K_1} = \frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

or, $\ln\left(\frac{0.24}{2\times10^{-3}\times60}\right) = \frac{E_a}{2}\left(\frac{1}{300} - \frac{1}{330}\right)$

$\therefore E_a = 4620 cal/mol = 4.62 kcal/mol.$

50. Ans (333)

$$P_A = X_A \cdot P^\circ_A = Y_A \cdot P_{total}$$

$$\therefore P^o_A = \frac{0.6\times500}{0.4} = 750 \text{ torr}$$

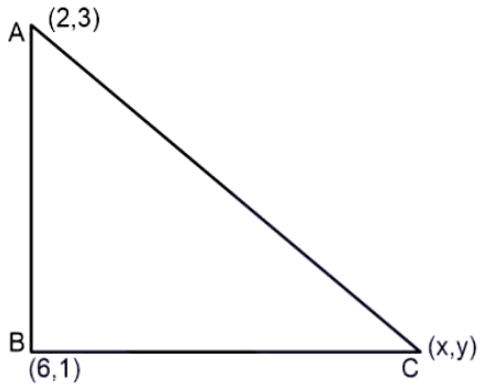
and $P^o_B = \frac{0.4\times500}{0.6} = 333.33$ torr.

MATHS

51. Ans (D)

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Slope (AB). Slope $(BC) = -1$

$$\Rightarrow \frac{1-3}{6-2} \cdot \frac{y-1}{x-6} = -1$$

$$\Rightarrow +\frac{2}{4} \cdot \left(\frac{y-1}{x-6}\right) = +1$$

$$\Rightarrow y-1 = 2(x-6) \Rightarrow 2x-y = 11 \#(1)$$

Area of

$$\triangle ABC = \frac{1}{2}\left|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right|$$

$$= \frac{1}{2}\left|2(1-y)+6(y-3)+x(3-1)\right| = 10\sqrt{2}$$

$$= \left|2x+4y-16\right| = 20\sqrt{2} \quad (2)$$

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Put $y = 2x-11$ in (2)

$$x = 6+2\sqrt{2}$$

$$\therefore y = 1+4\sqrt{2}$$

52. Ans (B)

$\vec{x} = (\vec{a}\times\vec{b})\times\vec{c}\,\vec{a}\cdot\vec{c} = 4$

$= (\vec{a}\cdot\vec{c})\vec{b} - (\vec{b}\cdot\vec{c})\vec{a}$

$\vec{b}\cdot\vec{c} = 2$

$\vec{x} = 4\vec{b} - 2\vec{a}$

$\vec{x} = 6\hat{i} - 8\hat{j} + 2\hat{k}$

$\Rightarrow$ Unit vector a long $\vec{x} = \frac{1}{2\sqrt{26}}\left(6\hat{i}-8\hat{j}+2\hat{k}\right)$

$\Rightarrow$ Projection of $\hat{x}$ on $\vec{a} = \left|\hat{x}\cdot\frac{\vec{a}}{|\vec{a}|}\right| = \frac{2}{\sqrt{39}}$

53. Ans (A)

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Lebel the 10 colinear points $1,2,\ldots\ldots,10$ let the chosen points be $i_1 < i_2 < i_1 < i_4$ with the condition $i_{k+1} - i_k \geqslant 2$

Define new variables

$j_1 = i_1, j_2 = i_2 - 1, j_3 = i_3 - 2; j4 = i_4 - 3$

Then $1 \leqslant j_1 \leqslant j_2 < j_3 < j_4 \leqslant 7$ so we have to just choose is distinct numbers from $\{1,2,7\}$

$\Rightarrow$ No. of days we $7_{C_4} = 35$.

54. Ans (B)

$f(x) = g^{-1}(x)$

$g(f(x)) = x$

$g'(f(x))f'(x) = 1$

$g'(f(x)) = \dfrac{1}{f'(x)}$

Putting $f(x) = 2\pi$

$2\pi = (2x - 3\pi)^5 + \dfrac{4}{3}x + \cos x$

$x = \dfrac{3\pi}{2}$ (only solution)

$\therefore f\left(\dfrac{3\pi}{2}\right) = 2\pi$

$\Rightarrow g'(2\pi) = \dfrac{1}{f'\left(\dfrac{3\pi}{2}\right)}$

$= \dfrac{1}{\left(\dfrac{7}{3}\right)}$

$= \dfrac{3}{7}$

55. . Ans (C)

$f(x) = \int_0^x e^{x-t}dt + \int_x^1 e^{t-x}dt$

$f(x) = e^x \int_0^x e^{-t}dt + e^{-x} \int_x^1 e^t dt$

$f(x) = e^x\left(1 - e^{-x}\right) + e^{-x}\left(e - e^x\right)$

$f(x) = x^x + e^{1-x} - 2$

$f'(x) = e^x - e^{1-x} = 0$

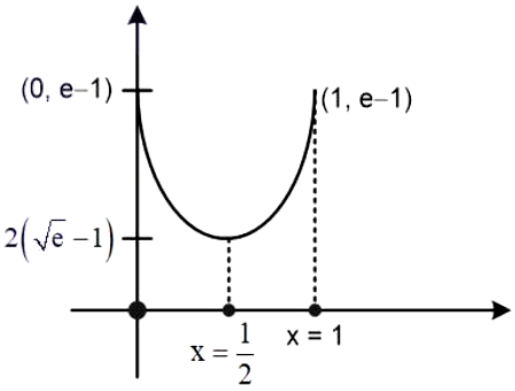
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$$x = 1 - x$$

$$x = \frac{1}{2}$$

$f\left(\frac{1}{2}\right) = 2\left(\sqrt{e} - 1\right) \rightarrow$ Minimum value



graph of $f(x)$ in [0,1]

56. Ans (C)

$$\frac{z}{1-z} = -1 + \frac{1}{1-z}$$

$$\Rightarrow s = \sum_{k=1}^{3}\left(-1 + \frac{1}{1-z_k}\right) = -3 + \sum_{k=1}^{3}\frac{1}{1-z_k}$$

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If $f(z) = (z-z_1)(z-z_2)(z-z_3)$

$$\Rightarrow \frac{f'(z)}{f(z)} = \sum_{k=1}^{3}\left(\frac{1}{z-z_k}\right)$$

Put $z = 1$

$$\sum_{k=1}^{3}\left(\frac{1}{1-z_k}\right) = \frac{f'(1)}{f(1)}$$

$\because f'(z) = 3z^2 - 3$

$$\Rightarrow \sum_{k=1}^{3}\left(\frac{1}{1-z_k}\right) = \frac{0}{1} = 0$$

$$\Rightarrow s = -3 + 0 = -3$$

57. Ans (C)

Let $A_n = \int_0^1 x^3\left(1-x^2\right)^n dx$

Put $x^2 = t$

$$xdx = \frac{dt}{2}$$

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$\therefore A_n = \frac{1}{2}\int_0^1 t(1-t)^n dt$

Using king property

$A_n = \frac{1}{2}\int_0^1 t^n (1-t)\,dt$

$\Rightarrow A_n = \frac{1}{2}\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$

$\therefore \sum^{\infty} A_n = 1$

58. Ans (A)

let $g'(1) = \alpha, g(2) = \beta$

$f(x) = x^2 + \alpha x + \beta$

$g(x) = (1+\alpha+\beta)x^2 + (2x+\alpha)x + 2$

$= (3+\alpha+\beta)x^2 + x\alpha + 2$

$g'(x) = 2(3+\alpha+\beta)x + \alpha, g''(x) = 2(3+\alpha+\beta)$

$\alpha = g'(1) = 2(3+\alpha+\beta) + \alpha$

$2\alpha + 2\beta = -6$ and $\alpha + \beta = -3$

$\beta = g''(2) = 2(3+\alpha+\beta), f''(3) = 2$

$2\alpha + \beta = -6$

$g''(2) = 0$

$\alpha = -3$ and $\beta = 0$

$f(x) = x^2 - 3x$

$\frac{1-f''(3)}{1-g''(2)} = \frac{1-2}{1-0} = -1$

$f'(x) = 2x - 3, f''(x) = 2$

59. Ans (A)

Given data set $\{8,9,10,11,12\}$

$\Rightarrow \bar{x} = \frac{\sum x_i}{5} = 10$

(A) New mean $= 2\times 10 = 20$ (Double of old mean)

(B) New data set $\{10,9,10,11,10\}$

$$\text{New mean} = \frac{10+9+10+11+10}{5} = 10$$

(C) Variance $(\sigma^2) = \frac{\sum (x_i)^2}{5} - \left(\frac{\sum x_i}{5}\right)^2$

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$$= \left\{ \frac{8^2 + 9^2 + 10^2 + 11^2 + 12^2}{5} \right\} - (10)^2$$

$$= 102 - 100$$

$$= 2$$

(D) New S.D. $= 2$ (Old S.D.)

$$= 2\sqrt{2}$$

$$= 2 \times (1.4142)$$

$$= 2.828$$

10. 60.Ans (B)

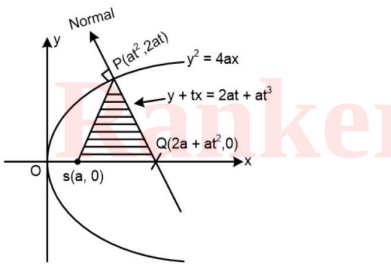
$7^{\sqrt{\log_7 5}} = 5^{\sqrt{\log_5 7}}$ and $5^{(\log_5 7)^{\frac{1}{3}}} = 7^{(\log_7 5)^{\frac{2}{3}}}$

$x^2 - 5x - 3 = 0;\ \alpha + \beta = 5; \alpha\beta = -3$

Now $\alpha + \frac{1}{\beta} = \frac{\alpha\beta + 1}{\beta} = -\frac{2}{\beta}$;

i.e. roots are $-\frac{2}{\alpha}; -\frac{2}{\beta}$.

61. Ans (D)



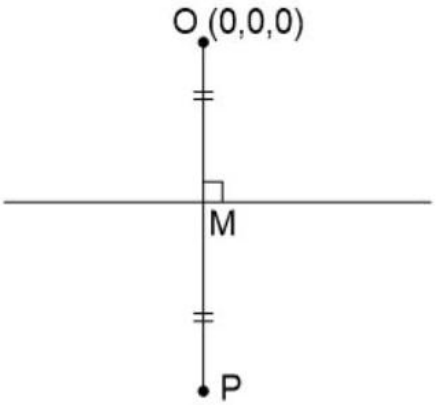
$$\text{Area (A)} = \frac{1}{2}\begin{vmatrix} at^2 & 2at & 1 \\ 2a + at^2 & 0 & 1 \\ a & 0 & 1 \end{vmatrix}, t = -m$$

$$= a^2\, m(1 + m)$$

62. Ans (C)

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$$\frac{x-1}{1}=\frac{x-2}{1}=\frac{z-3}{-1}=\lambda$$

Let $M\equiv(1+\lambda,2+\lambda,3-\lambda)$

$\overrightarrow{OM}=(1+\lambda)\hat{i}+(2+\lambda)\hat{j}+(3-\lambda)\hat{k}$

$\therefore OM\perp\left(\hat{i}+\hat{j}-\hat{k}\right)$

$\Rightarrow(1+\lambda)(1)+(2+\lambda)(1)-(3-\lambda)(+1)=0$

$\Rightarrow\lambda=0$

$\Rightarrow M\equiv(1,2,3)$

$\therefore PM=OM=\sqrt{1+4+9}=\sqrt{14}$

63. Ans (B)

$$=\sum_{i=0}^{p}\frac{m!}{i!(m-i)!}\cdot\frac{(n+i)!}{p!(n+i-p)!}\frac{(p+n)!}{(p-i)!(p+n-p+i)!}$$

$$=\sum_{i=0}^{P}\frac{m!}{i!(m-i)!}\frac{n!}{(n+i-p)!(p-i)!}\cdot\frac{(p+n)!}{p!n!}$$

$$={}^{n+p}C_n\sum_{i=0}^{P}{}^{m}C_i\,{}^{n}C_{p-i}$$

$${}^{n+p}C_n\cdot{}^{m+n}C_p$$

64. Ans (B)

If $xy<1$, then equation has no solution

If $xy=1$, then equation has no solution

If $xy>1\Rightarrow x^2y^2>1$

$$\Rightarrow\pi+\tan^{-1}\left(\frac{x^2+y^2}{1-x^2y^2}\right)=[xy]\frac{\pi}{4}$$

$$\Rightarrow\tan^{-1}\left(\frac{x^2+y^2}{(xy)^2-1}\right)=\pi\left(1-\frac{[xy]}{4}\right)$$

possible when $3\le xy<4$

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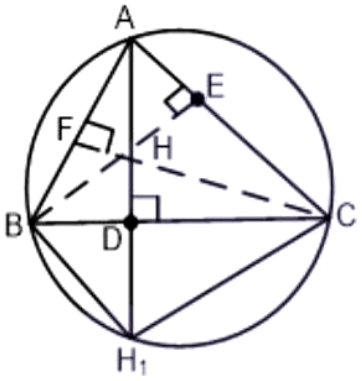
$\Rightarrow \frac{x^2+y^2}{x^2y^2-1}=1$ or $x^2+y^2=x^2y^2-1$

$\Rightarrow 9 \le x^2+y^2+1<16$

65. Ans (A)

Statement-1 : Apollonious theorem

Statement-2 :



$\angle CAD = \angle CBH_1 = 30^\circ - \angle C$

Also $\angle CBE = 30^\circ - \angle C$

$\Rightarrow \triangle HSD \simeq \triangle H_1BD$

$\Rightarrow HD = H, D$

66. Ans (C)

$\ell = 10+16+6$

67. Ans(A)

Put $e^t+e^{-t}=2\sec x, e^t-e^{-t}=2\tan x$

$\Rightarrow \left(e^t-e^{-t}\right)_{dt}=2\sec x\tan x dx$

$dt = \sec x dx$

$\therefore \int_0^{\log\left(1+\sqrt{2}\right)}\left(e^t+e^{-t}\right)^8 dt = \int_0^{\pi/4} 2^8(\sec x)^9 dx$

68. Ans (B)

Let roots are $\frac{a}{r}$, a , ar

$$a\left(1+r+\frac{1}{r}\right)=\frac{19}{2} \#(1)$$

$$a^2\left(1+r+\frac{1}{r}\right)=\frac{57}{2} \#(2)$$

$a^3 = 27 \Rightarrow a = 3$

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Solving $r=\frac{2}{3}$ or $\frac{3}{2}$ (rejected)

$\Rightarrow$ G.P. will be $\frac{9}{2},3,2,\ldots$

$S_\infty=\frac{27}{2}$

69. Ans (C)

$$\frac{1}{2}\frac{d\left(x^2+y^2\right)}{x^2+y^2}=\frac{xdy-ydx}{x^2+y^2}$$

$$=\frac{\frac{(xdy-ydx)}{x^2}}{\frac{x^2+y^2}{y^2}}=\frac{d\left(\frac{y}{x}\right)}{1+\left(\frac{y}{x}\right)^2}=\tan^{-1}\left(\frac{y}{x}\right)$$

70. Ans (A)

Let $A=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$

$AB=BA\Rightarrow b=c=f=0\ \&\ a=e=i$

$|A|=a^3=8\Rightarrow a=2$

$\Rightarrow \text{tr}(A)=3a=6$

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71. Ans (4)

Let A : drawn coin is unbiased.

B : Drawn coin is double - head.

C : drawn coin is double - tail

H : outcome of the toss is head

$P(A/H)$:- Probability that the coin is unbaised given that a head was observed.

Using Bayes theorem :-

$$P(A/H)=\frac{P(H/A)\cdot P(A)}{P(H)}$$

$$P(A)=\frac{15}{20}=\frac{3}{4};P(B)=\frac{2}{20}=\frac{1}{10},P(C)=\frac{3}{20}$$

$$P(H/A)=\frac{1}{2}$$

$$P(H/B)=1$$

$$P(H/C)=0$$

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$$\therefore P(H)=P(H/A)\cdot P(A)+P(H/B)\cdot P(B)+P(H/C)\cdot P(C)\Rightarrow\frac{19}{40}$$

$$\therefore P(A/H)=\frac{P(H/A)\cdot P(A)}{P(H)}=\frac{15}{19}$$

$\therefore$ Required probability is $\frac{m}{n}=\frac{15}{19}$

$$\therefore n-m=19-15=04$$

72. Ans (499)

$$I=\underbrace{\int\frac{\cos2x\cdot\cot3x}{\cot5x}dx}_{I_1}-\underbrace{\int\tan2x\tan5x\tan7xdx}_{I_2}$$

Again, $5x=2x+3x$

$$\cot5x=\cot(2x+3x)$$

$$\cot5x=\frac{\cot2x\cdot\cot3x-1}{\cot2x+\cot3x}$$

$$\cot5x\cdot\cot2x+\cot5x\cdot\cot3x+1=\cot2x\cdot\cot3x$$

$$I_1=\int\frac{\cot5x\cdot\cot2x+\cot5x\cdot\cot3x+1}{\cot5x}dx$$

$$I_1=\int\cot2x+\cot3x+\tan5xdx$$

$$I_2=\int\tan7x\cdot\tan5x\cdot\tan2xdx$$

$$I=I_1-I_2$$

$$I=\int(\cot2x+\cot3x+\tan5x)dx$$

$$-\int(\tan7x-\tan5x-\tan2x)dx$$

$$I=\int(\cot2x+\tan2x)dx+\int\cot3xdx$$

$$+2\int\tan5xdx-\int\tan7xdx$$

$$I=2\int\operatorname{cosec}4xdx+\int\cot3xdx+2\int\tan5xdx-\int\tan7xdx$$

$$I=\frac{1}{2}\ln(\operatorname{cosec}4x-\cot4x)+\frac{1}{3}\ln(\sin3x)+\frac{2}{5}\ln(\sec5x)+\frac{1}{7}\ln(\cos7x)+dx$$

$$I=\frac{1}{2}\ln(\tan2x)+\frac{1}{3}\ln(\sin3x)+\frac{2}{5}\ln(\sec5x)+\frac{1}{7}\ln(\cos7x)+C$$

Hence $a+b+c+d=\frac{1}{2}+\frac{1}{3}+\frac{2}{3}+\frac{1}{7}$

$$=\frac{289}{210}=\frac{m}{n}$$

$$m+n=289+210=499$$

73. Ans (2)

$$2^{(2^x-1)^2}+2^{2x}-2.2^x=\sqrt{x}+\log_2(1+\sqrt{x})$$

$$2^{(2^x-1)^2}+\underbrace{(2^x-1)^2}_{\text{concave up}}-1=\sqrt{x}+\underbrace{\log_2(1+\sqrt{x})}_{\text{concave down}}$$

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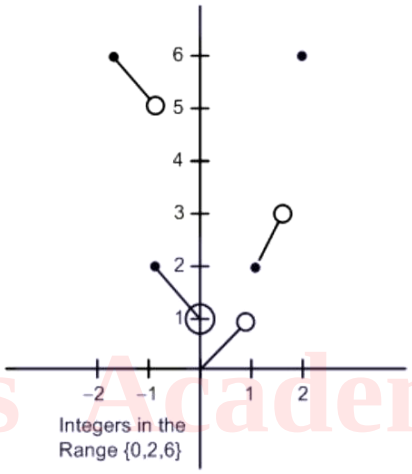
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74. Ans (7)

$f(x)=[x]^2+|x|$

$$=\begin{cases}[x]^2+x, & x\geqslant 0\\ [x]^2-x, & x<0\end{cases}$$

$$=\begin{cases}4-x & , \quad -2\leqslant x<-1\\ 1-x & , \quad -1\leqslant x<0\\ x & , \quad x<1\\ x+1 & , \quad 1\leqslant x<2\\ 6 & , \quad x=2\end{cases}$$



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for is discontinuous at $\{-1,0,1,2\}$

$m=3 \;\; m+n=7$

$n=4$

75. Ans (1)

$f(x)=\frac{x^3}{3}-\frac{x^2}{2}+1$

$f'(x)=x(x-1)$ and $f(0).f(1)>0$

$\Rightarrow f(x)=0$ has one root between

$x=-2$ and $x=-1$

$\Rightarrow f(f(x))=0$ and unique solution

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